

Computer Vision

Introduction

Technische Hochschule Rosenheim
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... used to be hard:



IN CS, IT CAN BE HARD TO EXPLAIN
THE DIFFERENCE BETWEEN THE EASY
AND THE VIRTUALLY IMPOSSIBLE.

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... and sometimes still is:



Chihuahua or Muffin?

© Twitter / Karen Zack / @teenybiscuit,
<https://twitter.com/teenybiscuit/status/707727863571582978>

Object Recognition ...



Cat or ice cream?

© Twitter / Karen Zack / @teenybiscuit,
<https://twitter.com/teenybiscuit/status/709538333219618816>



Sloth or chocolate roll?

© Twitter / Minh Le / @GoosemanCS,
<https://twitter.com/GoosemanCS/status/709589513832570880>

Main Course

1. Introduction
2. Image acquisition (camera sensors)
3. Camera calibration
4. Color
5. Image pre-processing: Threshold and filter operations
(histograms and thresholds, linear and non-linear filters)
6. Object classification: Convolutional Neural Networks (CNN)
7. Object localization and detection (R-CNN, YOLO)
8. Image segmentation - classic methods
(contour detection)
9. Image segmentation - with deep learning
(Mask R-CNN, U-Net/Autoencoder)
10. Stereo

AAI Add-On

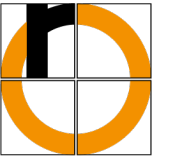
- Stereo: epipolar & fundamental matrix, 8-point algorithm
- Outlier elimination: RANSAC, LMedS
- Corner detection
- Feature Matching
- Structure-from-Motion

Especially recommended

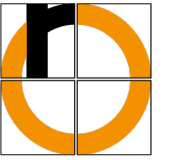
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<https://visionbook.mit.edu/>
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- Szeliski, R.: **Computer Vision: Algorithms and Applications**, Springer, 2nd edition, 2022.
- Goodfellow, I., Bengio, Y., Courville, A.: **Deep Learning**, MIT Press, 2017.
<http://www.deeplearningbook.org/>

Additional literature

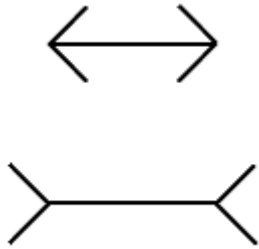
- A. Nischwitz, M.W. Fischer, G. Socher: **Computergrafik und Bildverarbeitung**, Band 2 Bildverarbeitung, Springer Vieweg, 4. Auflage, 2020.
- Aggarwal, Ch.C.: **Neural Networks and Deep Learning: A Textbook**, Springer, 2023.
- Hartley, R., Zisserman, A.: **Multiple View Geometry in Computer Vision**, Cambridge University Press, 2nd edition, 2004.



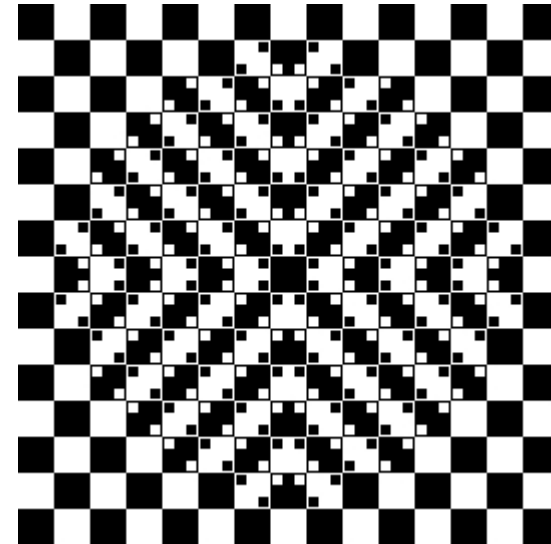
Human Vision: Optical Illusions



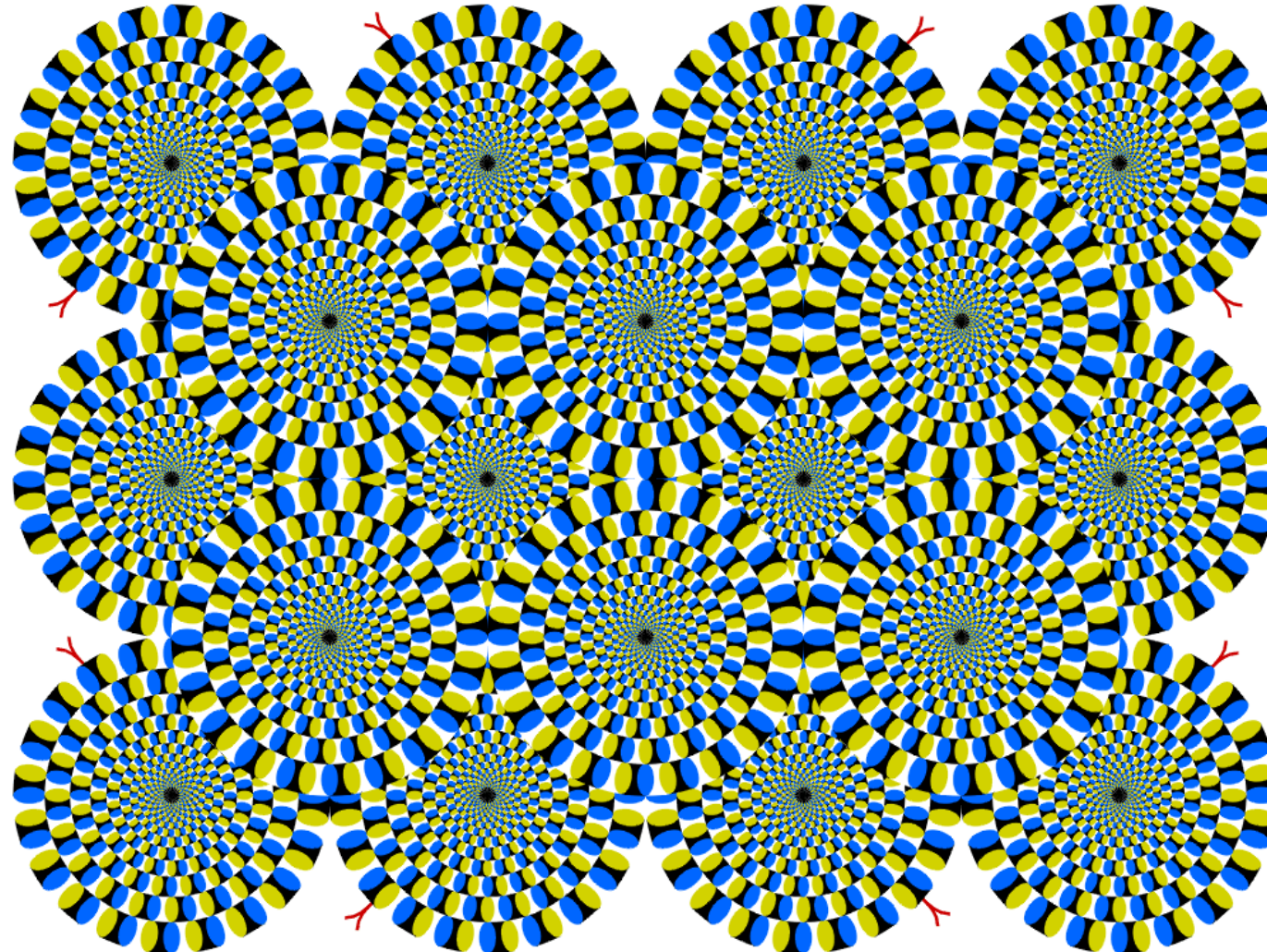
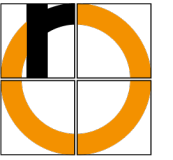
<http://www.ritsumeai.ac.jp/~akitaoka/index-e.html>



Are the two lines the same length?

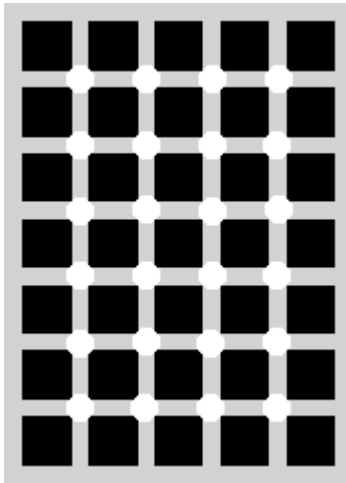


All fields are exact squares
All lines are exactly horizontal or vertical



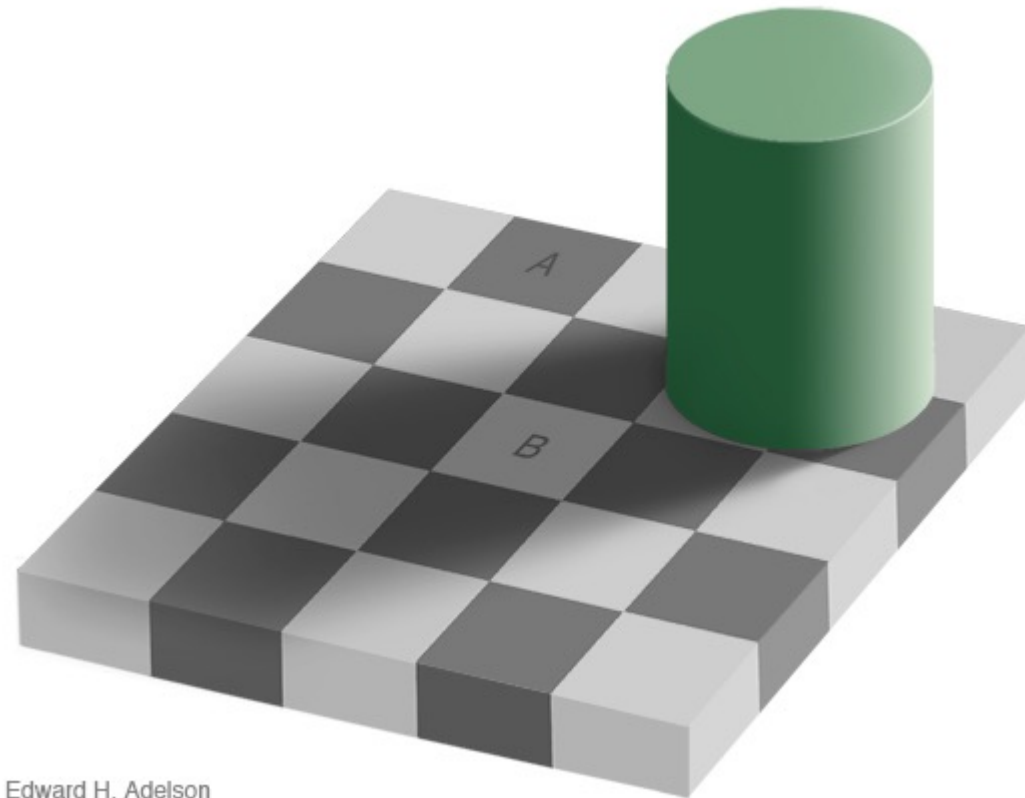
<http://www.ritsumei.ac.jp/~akitaoka/index-e.html>

http://www.cs.dartmouth.edu/~farid/Hany_Farid/Illusions/Entries/2011/6/6_Hermann_Grid.html



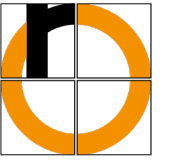
dark circles with
eye movement

http://web.mit.edu/persci/people/adelson/checkershadow_illusion.html



Edward H. Adelson

Field A and B have **exactly** the same brightness
(the same gray value: 120)



Terminology

- Image Processing
 - Images are transformed to images, often as a pre-processing step
- Image Analysis
 - Analytical processes that typically lead to a considerable reduction in data and a more compact description of the image
- Scene analysis and image understanding
 - Connection between recognized objects and the meaning for the user. Example: automatic navigation of autonomous vehicles. Artificial intelligence (AI) methods are often used here.
- Computer Vision
 - only vaguely defined
 - includes image and scene analysis as well as methods for 3D reconstruction and depth calculation (which can of course also play a role in scene analysis)

- Pattern Recognition

- Includes more than just image processing, e.g. also speech processing and other signals. In general: the processing of *patterns*.
- Classification of simple patterns
- Analysis of complex patterns

- Pattern

- an element of the problem area under consideration (e.g.: an image of a figure or an object)
- A pattern is a function

$$\mathbf{f}(\mathbf{x}) = \begin{pmatrix} f_1(x_1, x_2, \dots, x_n) \\ f_2(x_1, x_2, \dots, x_n) \\ \vdots \\ f_m(x_1, x_2, \dots, x_n) \end{pmatrix}$$

- Speech signal:
- Gray image:
- RGB color image:

$$f(t)$$

$$f(x, y)$$

$$\mathbf{f}(\mathbf{x}) = \begin{pmatrix} f_r(x, y) \\ f_g(x, y) \\ f_b(x, y) \end{pmatrix}$$

- Image sequence (video):

$$\mathbf{f}(\mathbf{x}) = \begin{pmatrix} f_r(x, y, t) \\ f_g(x, y, t) \\ f_b(x, y, t) \end{pmatrix}$$

- Simple pattern
 - Pattern can be classified as a whole
 - Class name is sufficient as a description
 - Examples:
Character recognition,
single spoken word
- Complex pattern
 - Classification as a whole pattern not possible
 - Class name not sufficient
 - Example:
Description of a street scene,
Interpretation of a CT scan

